

DPAssist for Teachers

Set up a cloud-based portfolio feedback system using Google Forms, Google Sheets, Streamlit Cloud, Render, Gmail, and Gemini.

No local Python required

Cloud dashboard

24/7 background worker

Teacher keeps final control

SETUP CHECKLIST

- 1 What you are setting up
- 2 Prepare Google Form and Sheet
- 3 Google Cloud and service account
- 4 Gmail App Password
- 5 Gemini API key
- 6 Deploy Streamlit dashboard
- 7 Deploy Render worker
- 8 Daily use and troubleshooting

1

What You Are Setting Up

DPAssist has two cloud services that work together:

Service	Platform	Purpose
Teacher Dashboard	Streamlit Cloud	Upload rubrics, view submissions, and check system status.
Background Worker	Render Starter Worker	Checks Google Sheets every 60 seconds, evaluates portfolios, sends emails, and updates statuses.
Data Store	Google Sheets	Stores student submissions, rubric data, and processing status.
AI Evaluation	Google Gemini	Evaluates portfolio evidence against teacher rubrics.

Important: Streamlit Cloud is the dashboard only. The Render background worker is required for automatic processing. If only the dashboard is deployed, submissions will appear in Google Sheets but will not be evaluated automatically.

Current production deployment has been verified with real student submissions using the Streamlit dashboard and Render worker.

Prepare the Google Form and Sheet

Students submit portfolio links through a Google Form. Google Forms automatically records each submission in a connected Google Sheet.

Required form information

- Student first name
- Student last name
- Student email address
- Class section, if applicable
- Unit name
- Published portfolio URL

Required Google Sheet columns

DPAssist is designed to work with Google Form response sheets that include these fields:

- Timestamp
- What is your class section?
- What is your last name?
- What is your first name?
- Select the unit
- A portfolio URL column with PUBLISHED in the column name
- What is your email address?

DPAssist automatically adds status-tracking columns if they do not already exist.

Unit names matter: The unit name in the rubric must match the unit name in the Google Form exactly. For example, Milling About is different from milling about or Milling About! .

Create Google Cloud Credentials

DPAssist needs secure access to the Google Sheet connected to your form.

Create a Google Cloud project

1. Go to <https://console.cloud.google.com>.
2. Sign in with your Google account.
3. Create a new project named **DPAssist**.
4. Make sure the DPAssist project is selected.

Enable APIs

1. Open **APIs & Services > Library**.
2. Enable **Google Sheets API**.
3. Enable **Google Drive API**.

Create a service account

1. Open **APIs & Services > Credentials**.
2. Choose **Create Credentials > Service Account**.
3. Name it **dpassist-reader**.
4. Skip optional role steps and finish creating the account.
5. Open the service account, go to **Keys**, and create a new **JSON** key.
6. Save the downloaded JSON file securely.

Do not upload the JSON file to GitHub. It contains private credentials. You will paste its contents into Streamlit Secrets and Render environment variables later.

Share the Google Sheet

1. Open the downloaded JSON file in a text editor.
2. Find the `client_email` value.
3. Open your Google Form responses Sheet.
4. Click **Share**.
5. Add the service account email as an **Editor**.
6. Uncheck **Notify people** and click **Share**.

Copy the Sheet ID

The Sheet ID is the long string between `/d/` and `/edit` in the Google Sheet URL.

```
https://docs.google.com/spreadsheets/d/THIS_PART_IS_THE_SHEET_ID/edit
```

4

Create a Gmail App Password

DPAssist sends student feedback emails through Gmail using an App Password, not your normal Google password.

1. Go to <https://myaccount.google.com>.
2. Open **Security**.
3. Make sure **2-Step Verification** is turned on.
4. Search for **App passwords**.
5. Create an app password named **DPAssist**.
6. Copy the 16-character password immediately.

Remove spaces. Google may display the app password in groups, such as `abcd efgh ijkl mnop`. When entering it in Streamlit or Render, use `abcdefghijklmnop` with no spaces.

5

Create a Gemini API Key

1. Go to <https://aistudio.google.com/app/apikey>.
2. Sign in with your Google account.
3. Click **Create API Key**.
4. Copy the key and store it securely.

The key is used by DPAssist to evaluate portfolios and parse rubric content.

Deploy the Streamlit Teacher Dashboard

The Streamlit app is the teacher-facing dashboard. It lets teachers upload rubrics, view submissions, and confirm system configuration.

1. Go to <https://share.streamlit.io> and sign in with GitHub.
2. Create a new app.
3. Select repository `bmorrow574/DPAssist` or your fork.
4. Set branch to `main`.
5. Set main file path to `portfolio_reviewer/teacher_ui.py`.
6. Open **Advanced settings** > **Secrets**.

Paste the following secrets, replacing placeholder values with your real values:

```
GOOGLE_SHEET_ID = "paste-your-sheet-id-here"
GEMINI_API_KEY = "paste-your-gemini-key-here"
TEACHER_EMAIL = "your.email@gmail.com"
GMAIL_APP_PASSWORD = "your16characterapppassword"
RUBRIC_STORAGE = "sheets"
CHECK_INTERVAL_SECONDS = "60"
GOOGLE_CREDENTIALS_JSON = '''
paste the entire contents of service-account-credentials.json here
'''
```

Triple quotes are required around the Google credentials JSON in Streamlit Secrets. Paste the entire JSON file, including the opening `{` and closing `}`.

Deploy the app. When the dashboard opens, go to **System Status** and confirm the required services show green check marks.

Deploy the Render Background Worker

The Render worker is what actually processes submissions. It must be deployed for DPAssist to run automatically.

Cost note: The working production deployment uses a Render Starter Worker. At the time of deployment, this was \$7/month. Render pricing may change.

1. Go to <https://render.com> and sign in with GitHub.
2. Select **New > Blueprint**.
3. Connect the GitHub repository.
4. Render should detect `render.yaml`.
5. Create a Blueprint instance, such as **DPAssist-Production**.
6. Set branch to `main`.
7. Leave Blueprint path as `render.yaml`.

Render environment variables

Enter these values when Render prompts for them:

Variable	Value
<code>GOOGLE_CREDENTIALS_JSON</code>	Entire contents of the service account JSON file
<code>GOOGLE_SHEET_ID</code>	Your Google Sheet ID
<code>GEMINI_API_KEY</code>	Your Gemini API key
<code>TEACHER_EMAIL</code>	The Gmail address that sends feedback
<code>EMAIL_APP_PASSWORD</code>	The Gmail App Password with no spaces

The following values are defined in `render.yaml`:

```
RUBRIC_STORAGE=sheets
CHECK_INTERVAL_SECONDS=60
```

Deploy the worker. In Render logs, successful startup should show that the service is live and running `python portfolio_reviewer/background_service.py`. After deployment, the worker checks the Google Sheet every 60 seconds.

Add a rubric

1. Open the Streamlit Teacher Dashboard.
2. Click **Rubrics**.
3. Upload the rubric PDF.
4. Enter the unit name exactly as it appears in the Google Form.
5. Set the due date.
6. Click **Parse and Add Rubric**.

Monitor submissions

Open **Submissions** to view student names, units, due dates, and status.

Status	Meaning
Feedback sent	Student feedback email was sent successfully.
Teacher draft created	A teacher-facing draft/review message was created for a late submission.
Portfolio access issue emailed	The student's portfolio requires sign-in or is not publicly accessible.
Processing	The worker is actively evaluating the submission.

Student portfolio access

Students must publish portfolio links so they are accessible without Google sign-in. DPAssist now detects Google sign-in pages and emails the student instructions to fix the access issue.

Teacher control

DPAssist provides feedback and draft evaluation language. Teachers remain responsible for reviewing student work, assigning final grades, and making instructional decisions.

Nothing is processing

- Confirm the Render worker is live.
- Check Render logs for errors.
- Confirm the Google Sheet ID is correct.
- Confirm the service account has Editor access to the Sheet.

Dashboard works, but submissions do not process

This usually means the Streamlit dashboard is deployed but the Render worker is not running.

Student did not receive feedback

- Check the student's email address in the Sheet.
- Confirm the Gmail App Password has no spaces.
- Check Render logs for SMTP errors.

Rubric not found

- Confirm the rubric was uploaded.
- Confirm the unit name matches exactly.
- Confirm `RUBRIC_STORAGE=sheets`.

Portfolio access issue

Ask the student to open the submitted link in an incognito/private window. If Google asks them to sign in, the site is not public yet.

Security Reminders

- Never commit `.env` to GitHub.
- Never commit `service-account-credentials.json` to GitHub.
- Store credentials only in Streamlit Secrets and Render Environment Variables.
- Use a Gmail App Password instead of a normal Google password.
- Remove spaces from the Gmail App Password before entering it in cloud services.

DPAssist - Cloud-based portfolio feedback for teachers

Teacher Dashboard: <https://dpassist-edu.streamlit.app>